

Day- 11 HDOC

Name - Aaryan Pratap Singh

Sap id - 590032842

Batch - 15

1. Algorithm

Step 1: Start.

Step 2: Declare variables `n`, `choice`, `temp`, `digit`, `sumSquares`, and `sumCubes`.

Step 3: Input a positive integer `n`.

Step 4: If `n <= 0`, display an invalid-input message and terminate.

Step 5: Display the menu:

1. Sum of squares of digits using **while loop**
2. Sum of cubes of digits using **for loop**
3. Difference between sum of cubes and sum of squares using **do-while loop**

Step 6: Input the user's `choice`.

Step 7: Use a `switch` statement to perform the selected operation.

Case 1 – Sum of Squares:

- Set `temp = n` and `sumSquares = 0`.
- Repeat while `temp > 0`:
 - `digit = temp % 10`
 - `sumSquares = sumSquares + digit × digit`
 - `temp = temp / 10`
- Display `sumSquares`.

Case 2 – Sum of Cubes:

- Set `sumCubes = 0`.
- Use a `for` loop with `temp = n`.
- For each digit:
 - `digit = temp % 10`
 - `sumCubes = sumCubes + digit × digit × digit`
 - `temp = temp / 10`
- Display `sumCubes`.

Case 3 – Difference:

- Set `temp = n`, `sumSquares = 0`, and `sumCubes = 0`.

- Using a **do-while** loop:
 - Extract the digit.
 - Add its square to **sumSquares**.
 - Add its cube to **sumCubes**.
 - Remove the digit.
- Calculate:
 $\text{Difference} = \text{sumCubes} - \text{sumSquares}$
- Display the difference.

Default: Display "Invalid choice!".

Step 8: Stop.

2. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Number = 123, Choice = 1	Sum of squares = 14	Sum of squares = 14	Pass
2	Number = 123, Choice = 2	Sum of cubes = 36	Sum of cubes = 36	Pass
3	Number = 123, Choice = 3	Difference = 22	Difference = 22	Pass
4	Number = 5, Choice = 1	Sum of squares = 25	Sum of squares = 25	Pass
5	Number = 10, Choice = 2	Sum of cubes = 1	Sum of cubes = 1	Pass
6	Number = 99, Choice = 3	Difference = 1296	Difference = 1296	Pass
7	Number = 123, Choice = 4	Invalid choice!	Invalid choice!	Pass
8	Number = 0	Invalid input	Invalid input	Pass

Sample Calculation for 123

- Sum of squares = $1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$
- Sum of cubes = $1^3 + 2^3 + 3^3 = 1 + 8 + 27 = 36$
- Difference = $36 - 14 = 22$

3. C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int n, choice, temp, digit;
```

```
long long sumSquares = 0, sumCubes = 0;
```

```
printf("Enter a positive integer: ");

scanf("%d", &n);

if (n <= 0)

{

    printf("Invalid input! Please enter a positive integer.\n");

    return 0;

}

printf("\nMENU\n");

printf("1. Sum of squares of digits (while loop)\n");

printf("2. Sum of cubes of digits (for loop)\n");

printf("3. Difference: sum of cubes - sum of squares (do-while loop)\n");

printf("Enter your choice: ");

scanf("%d", &choice);

switch (choice)

{

    case 1:

        temp = n;

        while (temp > 0)

        {

            digit = temp % 10;

            sumSquares = sumSquares + digit * digit;

            temp = temp / 10;

        }

        printf("Sum of squares of digits = %lld\n", sumSquares);

        break;
```

case 2:

```
for (temp = n; temp > 0; temp = temp / 10)

{

    digit = temp % 10;

    sumCubes = sumCubes + digit * digit * digit;

}

printf("Sum of cubes of digits = %lld\n", sumCubes);

break;
```

case 3:

```
temp = n;

do

{

    digit = temp % 10;

    sumSquares = sumSquares + digit * digit;

    sumCubes = sumCubes + digit * digit * digit;

    temp = temp / 10;

} while (temp > 0);

printf("Difference = %lld\n",

    sumCubes - sumSquares);

break;
```

default:

```
printf("Invalid choice!\n");
```

```
}
```

```
return 0;
```

```
}
```

```
UW PICO 6.09 File: day10.c Modified

#include <stdio.h>

int main()
{
    int n, choice, temp, digit;
    long long sumSquares = 0, sumCubes = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Invalid input! Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Sum of squares of digits (while loop)\n");
    printf("2. Sum of cubes of digits (for loop)\n");
    printf("3. Difference: sum of cubes - sum of squares (do-while loop)\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            temp = n;

            while (temp > 0)
            {
                digit = temp % 10;
                sumSquares = sumSquares + digit * digit;
                temp = temp / 10;
            }

            printf("Sum of squares of digits = %lld\n", sumSquares);
            break;

        case 2:
            for (temp = n; temp > 0; temp = temp / 10)
            {
                digit = temp % 10;
                sumCubes = sumCubes + digit * digit * digit;
            }

            printf("Sum of cubes of digits = %lld\n", sumCubes);
            break;

        case 3:
            temp = n;

            do
            {
                digit = temp % 10;

                sumSquares = sumSquares + digit * digit;
                sumCubes = sumCubes + digit * digit * digit;

                temp = temp / 10;
            } while (temp > 0);

            printf("Difference = %lld\n", sumCubes - sumSquares);
            break;

        default:
            printf("Invalid choice!\n");
    }

    return 0;
}
```

```
UW PICO 6.09 File: day10.c Modified

    {
        digit = temp % 10;
        sumSquares = sumSquares + digit * digit;
        temp = temp / 10;
    }

    printf("Sum of squares of digits = %lld\n", sumSquares);
    break;

    case 2:
        for (temp = n; temp > 0; temp = temp / 10)
        {
            digit = temp % 10;
            sumCubes = sumCubes + digit * digit * digit;
        }

        printf("Sum of cubes of digits = %lld\n", sumCubes);
        break;

    case 3:
        temp = n;

        do
        {
            digit = temp % 10;

            sumSquares = sumSquares + digit * digit;
            sumCubes = sumCubes + digit * digit * digit;

            temp = temp / 10;
        } while (temp > 0);

        printf("Difference = %lld\n",
               sumCubes - sumSquares);
        break;

    default:
        printf("Invalid choice!\n");
    }

    return 0;
}
```

4. Compilation And Execution

All 3 cases executed successfully

```
[aaryanrajput@aaryans-macbook ~ % nano day10.c
[aaryanrajput@aaryans-macbook ~ % clang day10.c -o day10.c
[aaryanrajput@aaryans-macbook ~ % ./day10.c
Enter a positive integer: 123

MENU
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference: sum of cubes - sum of squares (do-while loop)
Enter your choice: 1
Sum of squares of digits = 14
[aaryanrajput@aaryans-macbook ~ % ./day10.c
Enter a positive integer: 123

MENU
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference: sum of cubes - sum of squares (do-while loop)
Enter your choice: 2
Sum of cubes of digits = 36
[aaryanrajput@aaryans-macbook ~ % ./day10.c
Enter a positive integer: 123

MENU
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference: sum of cubes - sum of squares (do-while loop)
Enter your choice: 3
Difference = 22
aaryanrajput@aaryans-macbook ~ %
```

5. Testing Result

The program was **actually compiled and executed** against all 8 test cases above. All expected outputs matched the actual outputs.

Overall Result: 8/8 test cases passed. 

Conclusion: The menu-driven C program correctly uses **switch-case**, a **while loop** for the sum of squares, a **for loop** for the sum of cubes, and a **do-while loop** for calculating the difference.